KaxaNuk
Sharing knowledge

INVESTMENT BOOTCAMP

WRITE YOUR HYPOTHESIS AND BUILD YOUR BOOKS.



04 From your S03 signals to experiment 1: your hypothesis in the blueprint, then your historical books, with what was known each day.

IN THIS SESSION

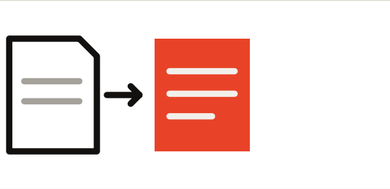
01 HYPOTHESIS

02 BUILD

03 REALISM

04 WRAP-UP

FROM YOUR SIGNALS TO YOUR BOOKS.

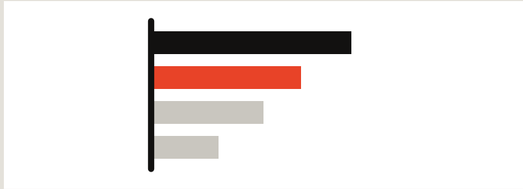


1 HYPOTHESIS

What you expect, before the rule.

- Experiment 1, with your signals
- Your hypothesis, refutable
- Drafted, criticized, saved

You leave with: your blueprint

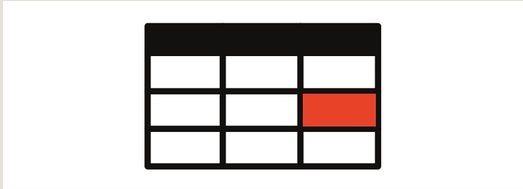


2 BUILD

Who, how much, when.

- Selection, sizing and timing
- From equal weight to fourteen methods
- Turnover: how often to rebalance

You leave with: your rule



3 REALISM

What was known that day.

- t – 1, and no survivors
- Corporate events: sell first
- Invariants and overfitting

You leave with: your books



4 WRAP-UP

What you learned, and what comes next.

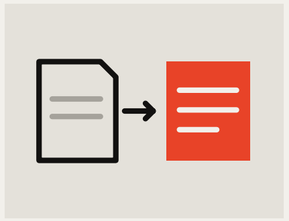
- The session's ideas, in your words
- Homework: your rule, your control, an alternative
- Next: the backtest, in S05

You leave with: your plan for S05

EIGHT STEPS. IN THIS SESSION, THE FOURTH.

S02 AND S03 | THIS SESSION | S05 | S06 | BEYOND THE COURSE

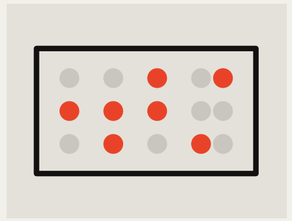
1 2 3 4 5 6 7 8



RESEARCH

S02

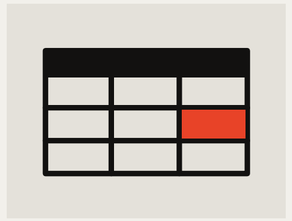
Your objective, from your library · OBJECTIVE.md



UNIVERSE

S02

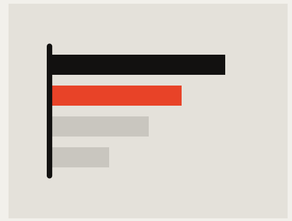
Which companies and when; delisted included · Universe/



DATA

S03

Your features, and whether they carry a signal · Data/

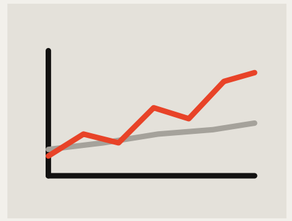


PORTFOLIO

S04

Your hypothesis, then the portfolios · Experiments/

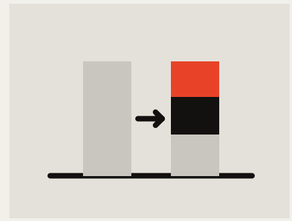
STEP 4



BACKTEST

S05

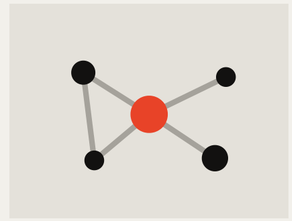
Your portfolio through time, net of costs



ATTRIBUTION

S05

Where the return came from



PAPER TRADING

S06

Tested without real money · Paper_Trading/



PRODUCTION

outside the course

From S03's homework

your kept signal, its table against holding, and N.

FROM YOUR SIGNALS TO EXPERIMENT 1.

ARYA · TREND-AND-VOLUME

Start experiment 1 with my kept signal. 1

I create Experiment_1/. First, your hypothesis: no rule yet. Go ahead?

go ahead

Done. Write what you expect and what would refute it. 2

EXPERIMENTS/EXPERIMENT_1/

`BLUEPRINT_1.md` your hypothesis, first

`JOURNAL_1.md` every decision, dated

`experiment_1.ipynb` the rule, later

`Portfolio/` each date's weights

```
Golden-Flow: the golden cross selects c_golden_cross
Golden-Flow: traded value weights r_traded_value_share
```

3

Golden-Flow · README.md

Own the KN US Equity Core stocks the market is trading most,

while their trend is up, in proportion to what they trade.

1. Your S03 signal becomes a hypothesis.
2. What you expect first; the rule after.
3. Golden-Flow: one signal selects, another weights.

`experiment-lifecycle`

WRITE IT DOWN BEFORE THE RULE.

2011–2019

2011

SIMMONS, NELSON & SIMONSOHN

DISCLOSURE

Undisclosed choices inflate false positives.

False-Positive Psychology

2014

GELMAN & LOKEN

FORKING PATHS

Data-dependent analysis: why many significant results fail.

The Statistical Crisis in Science

2017

BAILEY ET AL.

THE HOLDOUT

The usual hold-out tends to be unreliable.

The probability of backtest overfitting

2018

NOSEK, EBERSOLE, DEHAVEN & MELLOR

PREREGISTRATION

Plan the analysis before seeing the outcome.

The preregistration revolution

1

2019

ARNOTT, HARVEY & MARKOWITZ

YOUR FIRST PAPER

A holdout, opened once.

A Backtesting Protocol in the Era of Machine Learning

GOLDEN-FLOW NAMES ITS OWN

“The window is the choice”: a forking path, disclosed and counted.

THEREFORE

The save's time is the evidence.

Golden-Flow's line from its `BLUEPRINT_1.md`, written 2026-10-06.

NINE PARTS, DATED AND STAMPED.

GOLDEN-FLOW · BLUEPRINT_1.MD · OUTLINE

```
## Experiment 1 — …

**Written 2026-10-06, before any rule was coded.** Drafted with Researcher 0.31.0, a680873 (3)

### Thesis "a risk gate, not a return engine"

### The claim this moves Claim 1, the signal

### Rules universe, signal, sizing, cash, timing, lag, costs, control (1)

### What this experiment should show three predictions, five leads (2)

### Success criteria margins over its control, fixed now

### What would falsify it one condition; what may not rescue it

### Key risks "The pool is narrower than the book."

### Open questions this experiment deliberately does not answer the seven questions, skipped
```

1. *Rules*: the book in words; two cost rows, a cash reserve.
2. Predictions: each says where from.
3. The ninth: dated, stamped, before any rule.

The template's headings; Golden-Flow's lines quoted, or in brief, from its `BLUEPRINT_1.md`, written 2026-10-06 · KaxaNuk (2026).

A HYPOTHESIS YOU CAN REFUTE.

GOLDEN-FLOW'S HYPOTHESIS

the golden cross earns its place | +0.03 of Sharpe over its control

ITS CONTROL

the same rule, without the golden cross | on the rule's own dates

ITS FALSIFIER

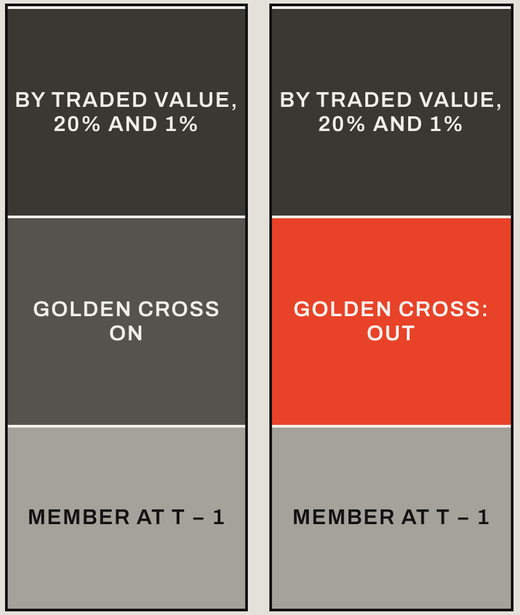
that prediction failing | no window or cap rescues it

YOUR HYPOTHESIS · TREND-AND-VOLUME

your S03 signal earns its place | against your control, without it

SCHEMATIC, NO DATA

Golden-Flow's rule and its control



The same rule, one ingredient out, on the same dates.

BEFORE LOOKING the margin, the control and the falsifier.

DRAFTED, CRITICIZED, SAVED.

ARYA · TREND-AND-VOLUME

/blueprint 1
1

SEVEN QUESTIONS, ONCE
Answer or skip; what you skip stays open.

Draft ready. I hand it to `blueprint-critic`, cold.
2

I answered every objection. Save it.

Saved, before any rule.
3

GOLDEN-FLOW · JOURNAL_1.MD · 2026-10-06

2026-10-06 — The blueprint's draft, put to its critic cold, ...

#	OBJECTION · 2 OF THE 12 BLOCKING	WHAT WAS DONE
1	“The falsifier and the claim table gave two verdicts ...”	“One verdict now: that book is falsified.”
3	“The control could hold names the book cannot ...”	“The control keeps the history requirement ...”

Twelve blocking, ten advisory: all answered before the commit.

1. Your draft, from your answers.
2. The critic reads it cold: objections only.
3. Saved: from now on, it does not change.

`blueprint` `blueprint-critic`

WHO, HOW MUCH AND WHEN.

THE QUESTION	WHAT IT DECIDES	GOLDEN-FLOW
Selection: who	Which stocks may be in the book on each date.	KN US Equity Core members with the golden cross on at t – 1.
Sizing: how much	How much of each, and how much in cash.	By their traded value, between 1% and 20%.
Timing: when	How often you change your mind: by calendar or by a signal.	Its dates and its trigger, from the blueprint; and when a stock leaves.

ONE AT A TIME change one question between two books, or you will not know which mattered.

FROM EQUAL WEIGHT TO FOURTEEN METHODS.

FAMILY	METHODS	WHAT FOR
Equal	Equal weight.	The one to beat: it needs nothing else.
By a feature	Proportional to a feature; the KN Index.	What trades most, or is largest, weighs more.
By risk	Inverse volatility; risk parity; maximum diversification.	The calmer ones weigh more.
Optimized	Mean-variance; Black-Litterman; constrained mean-variance.	The best estimated balance: sensitive to errors.
By clusters	HRP, HERC, NCO.	Groups the similar ones and splits across groups.
By losses	CVaR, CDaR.	Limits the worst falls.

CHOOSE BY concentration, turnover and capacity; never by its Sharpe in the window.

SAME UNIVERSE, ANOTHER SIZE.

EVERY MONTH END	EQUAL WEIGHT	TRADED VALUE	INVERSE VOLATILITY
Annual return	6.3%	7.0%	6.4%
Volatility	23.0%	25.5%	19.3%
Sharpe	0.30	0.32	0.32
Max drawdown	−73.6%	−82.3%	−63.1%
Weight of the top 10	20%	47%	32%

OUR CALCULATION · SHARADAR the 50 most traded with the 50/200 cross, 2000 to 2026, no costs.

HOW OFTEN? CALENDAR OR SIGNAL.

BY TRADED VALUE	EVERY QUARTER	EVERY MONTH	WHEN THE SIGNAL CHANGES
Rebalances a year	4	12	144
Turnover a year	161%	242%	441%
Sharpe, no costs	0.27	0.32	0.37
Max drawdown	−87.6%	−82.3%	−76.6%

FASTER COSTS MORE almost double the turnover for a little Sharpe; S05 charges it.

CONSTRAINTS: LEVERS, NOT ORNAMENTS.

Your limits

Written in the blueprint, before building.

A cap

None above a fifth: concentration.

A floor

None too small: it must matter and be tradable.

A minimum count

If there are not enough, invest less.

Cash

A real instrument, like BIL: the rest goes there.

What a cap frees goes to cash, unless the blueprint says otherwise; one lever at a time.

TWO TRACKS, ONE BOOK.

INVESTMENT LAB

NO CODE

- your universe and your signal →
- a sizing method →
- rebalancing: calendar or signal →
- cap, floor, cash

Sizing: equal weight, risk parity, HRP…

Timing: month end, quarter or signal.

ARYA · TREND-AND-VOLUME

Write the rule from my blueprint. 1

Cross at t – 1, by traded value with a cap and a floor, month end, BIL. Go ahead?

go ahead

Section 2 written. Read the diff: is it what your blueprint says? 2

1. Code: the rule, from your blueprint.
2. Read the diff against the blueprint.

`portfolio-construction-runs`

WHAT WAS KNOWN THAT DAY.

EXPERIMENT_1.IPYNB · SECTION 2 · TWO DATES, INVENTED

```
rebalance 2024-04-30 · t
signal, membership, traded value: the close of 2024-04-29 · t - 1   1
bought: at the price of 2024-04-30 · t                              2
inverse volatility: returns to 2024-04-29 · one allocator per date
re-strike 2024-05-14 · t
EXR: last price on 2024-05-14 → sold on 2024-05-14                  3
the same day of foresight for every book compared
```

1. Read at t – 1.
2. Bought at t.
3. No price tomorrow? Sold today.

Leak
a history that includes the day it weights.

NO SURVIVORS.

Your universe, on each date

Who was a member that day.

From its entry

None before the date it joined.

To its exit

None after it left the benchmark.

The dead

Bought, merged or bankrupt: in while they existed.

The proof

Of the 643 members of 1998, more than half no longer trade.

Today's universe, applied to the past, holds only winners: the backtest discovers the market goes up.

CORPORATE EVENTS: SELL FIRST.

TWO ACQUISITIONS · KN US EQUITY CORE · SHARADAR

```
Wyeth · bought by Pfizer
review 2009-09-18: a member, 0.58% of the benchmark
last price: 2009-10-15                                  1
your book: sold on 2009-10-15, into cash                2
Genentech · bought by Roche
review 2009-03-20: a member, 1.14% of the benchmark
last price: 2009-03-26 · sold that day
held after it, the engine does not run                  3
```

1. The series stops: the deal closed.
2. Sold on its last priced day.
3. If not, the backtest does not run.

Almost always known

a deal is announced months ahead; a suspension without warning is the rare case.

EVERY COLUMN SUMS TO ONE.

PORTFOLIO/PORTFOLIO_WEIGHTS.CSV

TICKER	2024-04-30	2024-05-31	2024-06-28	
EXF	0.25	0.50	0	1
EXG	0.25	0	0	
EXH	0.25	0.50	0	
EXJ	0.25	0	0	2
BIL	0	0	1.00	3

EXPERIMENT_1/

Portfolio/ ignored by git

portfolio_weights.csv rebuilt by 3.1

experiment_1.ipynb commit this

JOURNAL_1.md

```
+ - Weight files: rule → portfolio_weights.csv · control → … · inverse volatility → …
```

1. `Ticker` first; one date per column.
2. Each column sums to exactly 1.
3. Nothing eligible: all in `BIL`.

Teaching computation, invented names.

Cash
a real, priced instrument: the weight file has no cash row.

FIVE INVARIANTS, EVERY BOOK.

EXPERIMENT_1.IPYNB · 2.1 AND 8

2.1 · The invariants, on every book

INVARIANT	WHAT IT CATCHES
✓ at most fully invested	more than fully invested, or negatively
✓ no negative weight	a short in a long-only book
✓ signal on the prior close	a signal not on the prior close 1
✓ a fill price	a name with no fill price
✓ nothing held past its end	a name held after it stopped trading

ARYA · TREND-AND-VOLUME

Bring my trial count up to date.

Your S03 analysis and the rule's book. Save? 2

Yes.

JOURNAL_1.md

```
+ ## <date> — The rule's book, built
+ - Outcome / decision: 2.1 passed. Trials so far: 2.
```

1. Check the signal itself.
2. Count every book once.
3. Verify raises when wrong.

HOW NOT TO OVERFIT.

THE PRACTICE	WHY
The blueprint, before the rule	What is written after proves nothing.
A control, one ingredient out	Only then do you know which part did it.
Count N: every book	Trying many and keeping the best inflates the result.
A holdout, opened once	The test you did not see while deciding.
The method, by a property	Never by its Sharpe in the window that judges it.

ONE LEVER AT A TIME each earns its place against the book without it.

THE SESSION'S IDEAS, IN YOUR WORDS.

01

Why the hypothesis before the rule?

02

Why equal weight first?

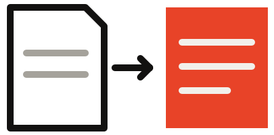
03

Rebalance at t: when do you read the signal and when do you buy?

04

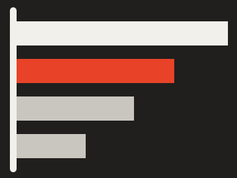
A stock bought by another company: what does your book do?

BEFORE S05: BUILD AND COUNT.



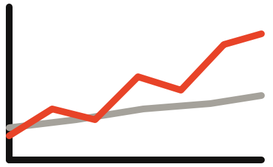
01 YOUR HYPOTHESIS

- Your blueprint, criticized and saved.
- Your control: one ingredient out.
- Ask `next`.



02 YOUR BOOKS

- Your rule, your control, an alternative.
- The five invariants, passing.
- Their turnover, measured.



03 READY FOR S05

- N up to date in your journal.
- Golden-Flow: its weight files.
- Your two cost rows.
- First Steps: FM P05, MF P03, MF P04.

PREDICTED FIRST. BUILT NEXT.

Your hypothesis, saved before the rule; your books, with what was known each day.

LEARN

The Bootcamp continues, then the pillars.

`github.com/KaxaNuk/Investment-Bootcamp`

TEST

Your strategy on the Lab; universes and factors from Analytics.

`kaxanuk.mx/lab · kaxanuk.mx/analytics`

TELL US

What the critic found; what you changed.

QUESTIONS `workshops@kaxanuk.mx`

NEXT

Session 05 · Simulate Your Books and Attribute Their Return

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Tools read 2026-10-13; works and links, 2026-10-11; the figures, from Sharadar prices of the KN US Equity Core, whose methodology is still provisional, computed 2026-10-17, with no costs. Shared under CC BY 4.0; the KaxaNuk name, logo and isotipo are not.